

Tugas Lab. Algo Pertemuan 8

Nama : Fathurrohman

NPM : 252310040

Kelas : TI-25-PA2

Github: <https://github.com/Fathur123-TI/LabAlgoritmaSemester2.git>

1.

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  const int MAX = 10;
6
7  string antrean[MAX];
8  int depan = 0;
9  int belakang = -1;
10 int jumlah = 0;
11
12 bool isEmpty() {
13     return jumlah == 0;
14 }
15
16 bool isFull() {
17     return jumlah == MAX;
18 }
19
20 void enqueue() {
21     if (isFull()) {
22         cout << "Antrean penuh!\n";
23         return;
24     }
25
26     string nama;
27     cin.ignore();
28     cout << "Masukkan nama pengunjung : ";
29     getline(cin, nama);
30
31     belakang++;
32     antrean[belakang] = nama;
33     jumlah++;
34
35     cout << nama << " berhasil mengambil nomor antrean.\n";
36 }
37
38 void dequeue() {
39     if (isEmpty()) {
40         cout << "Antrean kosong!\n";
41         return;
42     }
43
44     cout << antrean[depan] << " dipanggil ke loket.\n";
45
46     for (int i = depan; i < belakang; i++) {
47         antrean[i] = antrean[i + 1];
48     }
49
50     belakang--;
51     jumlah--;
52 }
53
54 void peek() {
55     if (isEmpty()) {
56         cout << "Antrean kosong!\n";
57     } else {
58         cout << "Antrean terdepan : " << antrean[depan] << endl;
59     }
60 }
61
62 void size() {
63     cout << "Jumlah antrean : " << jumlah << endl;
64 }
65
66 void tampilSemua() {
67     if (isEmpty()) {
68         cout << "Antrean kosong!\n";
69         return;
70     }
```

```

70     }
71 }
72 cout << "Daftar Antrean:\n";
73 for (int i = 0; i < jumlah; i++) {
74     cout << i + 1 << ". " << antrean[i] << endl;
75 }
76 }
77
78 void cekKosong() {
79     if (isEmpty())
80         cout << "Antrean kosong.\n";
81     else
82         cout << "Antrean tidak kosong.\n";
83 }
84
85 void cekPenuh() {
86     if (isFull())
87         cout << "Antrean penuh.\n";
88     else
89         cout << "Antrean belum penuh.\n";
90 }
91
92 void menu() {
93     cout << "=== SISTEM ANTREAN LOKET PELAYANAN ===\n";
94     cout << "1. Ambil Antrean (Enqueue)\n";
95     cout << "2. Panggil Antrean (Dequeue)\n";
96     cout << "3. Lihat Antrean Terdepan (Peek)\n";
97     cout << "4. Jumlah Antrean (Size)\n";
98     cout << "5. Tampilkan Semua Antrean\n";
99     cout << "6. Cek Antrean Kosong\n";
100    cout << "7. Cek Antrean Penuh\n";
101    cout << "8. Keluar\n";
102 }
103
104 int main() {
105     int pilih;

```

```

107     menu(); // Menu hanya tampil sekali
108
109     do {
110         cout << "\nPilihan : ";
111         cin >> pilih;
112
113         switch (pilih) {
114             case 1:
115                 enqueue();
116                 break;
117
118             case 2:
119                 dequeue();
120                 break;
121
122             case 3:
123                 peek();
124                 break;
125
126             case 4:
127                 size();
128                 break;
129
130             case 5:
131                 tampilSemua();
132                 break;
133
134             case 6:
135                 cekKosong();
136                 break;
137
138             case 7:
139                 cekPenuh();
140                 break;

```

```

140                 break;
141
142             case 8:
143                 cout << "Terima kasih.\n";
144                 break;
145
146             default:
147                 cout << "Pilihan tidak valid!\n";
148         }
149     } while (pilih != 0);
150
151     return 0;
152 }
153

```

Output:

```
=== SISTEM ANTREAN LOKET PELAYANAN ===
1. Ambil Antrean (Enqueue)
2. Panggil Antrean (Dequeue)
3. Lihat Antrean Terdepan (Peek)
4. Jumlah Antrean (Size)
5. Tampilkan Semua Antrean
6. Cek Antrean Kosong
7. Cek Antrean Penuh
0. Keluar

Pilihan : 1
Masukkan nama pengunjung : fathur
fathur berhasil mengambil nomor antrean.

Pilihan : 1
Masukkan nama pengunjung : rehan
rehan berhasil mengambil nomor antrean.

Pilihan : 5
Daftar Antrean:
1. fathur
2. rehan

Pilihan : 2
fathur dipanggil ke loket.

Pilihan :
0
Terima kasih.

-----
Process exited after 127.4 seconds with return value 0
Press any key to continue . . . |
```